

Programming Using C++ Level 1

Structures



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Definition

- ◆ A structure (`struct`) is a user-defined data type in C++ that allows you to group related variables of different data types under one name.
- ◆ A structure can contain variables such as `string`, `int`, `float`, `char`, and `bool`.

Explanation

- ◆ This program defines two structures, ` StAddress` and ` StName` , to organize information about a person.
- ◆ ` StAddress` stores address information.
- ◆ ` StName` stores the person's first name, last name, and an ` StAddress` object.
- ◆ This creates a structure inside another structure.

Steps in main()

1 Define the `StAddress` structure

◆ The structure contains:

■ `Street`

■ `PostNumber`

■ `City`

■ `Country`

2 Define the `StName` structure

◆ The structure contains:

■ `FirstName`

■ `LastName`

■ `Address`

◆ `Address` is an object of type `StAddress`.

3 Create the variables

◆ The program creates a `StName` object called `Name`.

◆ It also creates variables for age, salary, gender, and marital status.

4 Read the user's information

◆ The program uses `getline()` to read text values such as names and address information.

◆ It uses `cin` to read numerical and character values.

■ Steps in main()

5. Use `cin.ignore()`

- ◆ `cin.ignore()` removes the remaining newline character from the input buffer after reading `Age`.

- ◆ This allows the following `getline()` to work correctly.

6. Access structure members

- ◆ Structure members are accessed using the dot `.` operator.

- ◆ For example:

Name.FirstName

- ◆ The nested address members are accessed using:

Name.Address.Street

7. Calculate the yearly salary

- ◆ The program calculates the yearly salary by multiplying the monthly salary by 12.

float YearlySalary = MonthlySalary * 12;

8. Display the information

- ◆ The program prints the person's information using the values stored in the variables and structures.

9. Display the marital status

- ◆ The program uses the conditional operator `?:` to display either `Yes` or `No`.

💡 Example

◆ If the user enters information for a person, the program stores the information inside the `Name` structure and its nested `Address` structure.

◆ For example:

 Name.FirstName

 Name.LastName

 Name.Address.Street

 Name.Address.City

➡ Result

◆ The program displays the complete information, including the person's name, age, address, salary, gender, and marital status.

Note

- ◆ A structure is defined using the `struct` keyword.
- ◆ The variables inside a structure are called members.
- ◆ The dot `.` operator is used to access structure members.

⊘ Warning

- ◆ ``cin >>`` and ``getline()`` behave differently when reading input.
- ◆ After using ``cin >>`` to read a value, ``cin.ignore()`` may be needed before using ``getline()`` to prevent the leftover newline from being read as an empty line.



Comparison

▨ Normal Variables

- ✓ Each variable is declared separately.
- ✓ Related information is not grouped together.
- ✓ Example: ``string FirstName;``

▤ Structures

- ✓ Related variables are grouped under one type.
- ✓ Different data types can be stored together.
- ✓ Example: ``StName Name;``

✦✦ Summary

- ✓ A structure groups related variables under one user-defined type.
- ✓ Structure members are accessed using the dot `.` operator.
- ✓ Structures can contain other structures, allowing related data to be organized in a nested way.

Thank you 😊